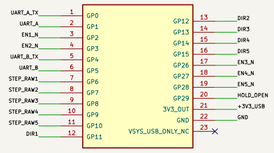
USB CONTROL / native USB-C on Waveshare RP2040-Zero



USB powers MCU and driver VIO only. VSYS/VBUS carrier pin is not connected.

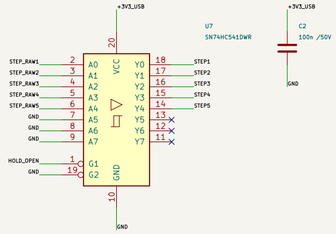
Existing CNC shield. Uno and their USB/supply remain completely unchanged.

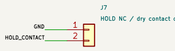


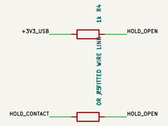


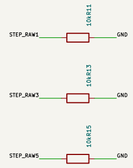
UART A: slots 1–4, addresses 0.1.2.3. UART B: slot 5, address 0.

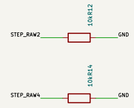
HOLD / hardware STEP suppression; software latches and waits for fresh ARM











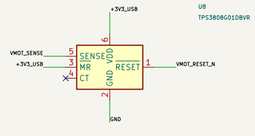
NC closed permits STEP; open/disconnected suppresses STEP. Not an emergency stop.

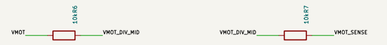
HOLD does not command torque off. USB or supply loss removes torque; support loads.

Firmware must stop/flush pulse queues and require a new ARM after any HOLD/fault.

R5: fit a clipped resistor-lead wire link, not a resistor. Dry contact only: no external voltage.

MOTOR-RAIL RESET / required separate two-wire RUN harness





Nominal failing trip 8.505 V. CT left open: reset release 12–28 ms.



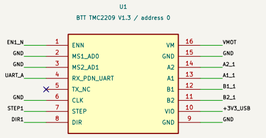
J9.1 -> Zero Key2.1 RUN; J9.2 -> Key2.2 GND. Harness is required.

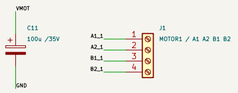
RUN is not on the 23 edge headers. Do not infer switch-pad left/right.

Reset supervision is not a latching E-stop; fast dips need physical testing.

With RUN harness fitted and VMOT absent, MCU stays reset; USB serial is unavailable.

SLOT 1 / NEMA 23



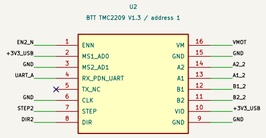


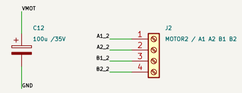




Current set via verified UART; frame size alone is not a current rating.

SLOT 2 / NEMA 23



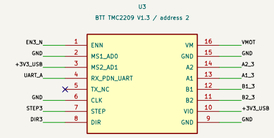


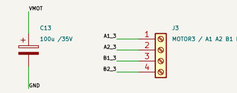




Current set via verified UART; frame size alone is not a current rating.

SLOT 3 / NEMA 17 / pancake



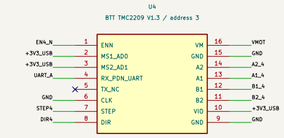






Current set via verified UART; frame size alone is not a current rating.

SLOT 4 / NEMA 17 / pancake



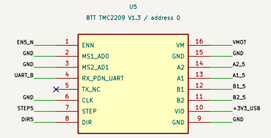


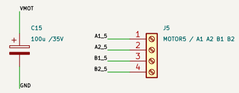




Current set via verified UART; frame size alone is not a current rating.

SLOT 5 / NEMA 17 / pancake



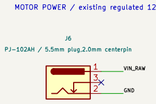






Current set via verified UART; frame size alone is not a current rating.

MOTOR POWER / existing regulated 12 V, 4 A supply. Check plug polarity before connecting.







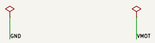












Initial total input budget <=2.5 A / 30 W. 3 A continuous only after testing; not five axes at maximum current.

Power paths must tolerate 4 A fault duration. D1 band/cathode faces VMOT: thermal test required.

Do not plug/unplug motors or driver modules while powered. TVS is not a continuous regeneration brake.

Unknown supply model, motor loads, cooling and transient energy prevent production clearance.

Required J9 to Zero RUN/GND harness; no edge-header RUN pad.
12 V: initial input 2.5 A / 30 W. 3 A after testing.

Sheet: /
File: big motor controller.kicad_sch

Title: Seizo / five-axis USB controller with motor-rail reset

Size: A2	Date: 2026-11-07	**Rev: R4 / PROTOTYPE**
KiCad E.D.A. 10.0.7		Id: 1/1